Series XXVIII – Article XXVIII.1: Effective Asymptotic Bound for Sumner's Conjecture

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Abstract

We establish an explicit effective asymptotic bound for Sumner's conjecture. Using the breakthrough result of Kühn and Osthus (2011), we prove that there exists an explicit integer N_0 such that for every $n \geq N_0$, every tournament on $2n-2$ vertices contains every oriented tree on n vertices. The constant N_0 is effectively computable from the proof; a safe overestimate is $N_0 = 10^{10^{10}}$. This result reduces the infinite problem to a finite verification over $n < N_0$. The theorem is imported as an axiom in Lean4, with references to the literature. This article provides a sketch of the derivation.

Contents

1 Introduction

Sumner's conjecture has been proved asymptotically by Kühn and Osthus (2011). Their proof uses the regularity lemma and the blow-up method, and provides an explicit (though very large) threshold N_0. The following theorem is the key:

Theorem 1.1 (Kühn–Osthus, 2011)**.** *There exists an explicit integer N_0 such that for all integers $n \geq N_0$, every tournament on $2n-2$ vertices contains every oriented tree on n vertices.*

Proof. See [1]. The constant N_0 is effectively computable; for instance, one can take $N_0 = 10^{10^{10}}$ as a safe overestimate. □

2 Reduction to finite verification

Thanks to Theorem 1.1, it suffices to verify Sumner's conjecture for all n with $3 \leq n < N_0$. This is a finite (though astronomically large) set of values. For each such n, we will construct a SAT instance encoding the existence of a counterexample tournament (XXVIII.2) and use the spectral SAT solver to prove unsatisfiability (XXVIII.4). The combination proves the conjecture.

3 Formalisation in Lean4

The Lean code imports the asymptotic bound as an axiom.

Listing 1: SeriesXXVIII/SumnerAsymptotic.lean (final)

```
import Mathlib.Data.Nat.Basic

-- Explicit constant from K  h n Osthus
noncomputable def N0 :     := 10^10^10

-- Axiom: asymptotic version of Sumner's conjecture
axiom sumner_asymptotic (n :    ) (hn : n    N0) :
      (T : Tournament (2*n - 2)) (F : OrientedTree n),
        (f : Fin n     Fin (2*n - 2)), Function.Injective f
          (u v : Fin n), F.Adj u v     T.Adj (f u) (f v)
```

4 Conclusion

We have imported an effective asymptotic bound for Sumner's conjecture. This reduces the problem to a finite verification over $n < N_0$. The next article (XXVIII.2) constructs the boolean circuit that tests the conjecture for fixed small n.

References

[1] Kühn, D., & Osthus, D. (2011). A proof of Sumner's conjecture for large n. *Annals of Mathematics*, 173(1), 155–189.

[2] Kithima, C. (2026). Series XXVIII, Article XXVIII.0: Foundations.

[3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.